

Harshitha Puttaswamy

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PROFESSIONAL SUMMARY

- Skilled Software Engineer with experience in full-stack development, backend engineering, and AI-driven applications
- Proficient in Python, TypeScript, and C, with a strong focus on ML, cloud computing, and reinforcement learning,
- Experienced in developing scalable applications like Splitwise-inspired expense tracker, and Autokube for Kubernetes automation
- Strong research background in Multi-Armed Bandits, Deep Q-Learning, and LLMs, with a proven track record in backend development (Django, MERN Stack, PostgreSQL, REST APIs), database optimization, and CI/CD pipelines (GitHub Actions, Docker, Kubernetes)

PROFESSIONAL EXPERIENCE

Research Assistant, Analysis of Envy-Freeness of Allocation Algorithms Using LLMs *February 2024 - Present*
Binghamton University, Department of Computer Science, Binghamton, NY, USA

- Developed and implemented a framework to analyze the envy-freeness of item allocations for n agents using different algorithms, evaluating responses with large language models (LLMs)
- Engineered prompt engineering techniques to assess agent preferences by asking LLMs whether agents would swap allocations, and used chi-square tests to evaluate the independence of responses across algorithms
- Investigating the impact of agent personality (based on the Big Five traits) on allocation preferences through statistical testing, with future plans to compare LLM results against MTurk data

Research Assistant, Iterative Voting for Committee Selection with Multi-Armed Bandits *January 2024 - December 2024*
Binghamton University, Department of Computer Science, Binghamton, NY, USA

- Developed and implemented a dynamic Multi-Armed Bandit (MAB) framework for iterative decision-making, optimizing voter utilities, and drawing parallels to complex bidding strategies in ad platforms as part of the Iterative Voting for Committee Selection with Multi-Armed Bandits project
- Engineered scalable algorithms for strategic optimization in multi-agent systems, leveraging Integer Linear Programming (ILP) to solve resource allocation and ranking problems akin to budget distribution
- Conducted extensive experimental analysis on preference aggregation and rule convergence, producing insights on maximizing utility and satisfaction, applicable to auction mechanisms and real-time bidding environments

Fyle Technologies Ltd, Backend Developer, Internship | Bangalore, India *March 2022 - December 2022*

- Developed and optimized APIs using Django, Angular, and PostgreSQL (PSQL), contributing to 8 new features, improving the product's market appeal
- Enhanced code quality by introducing unit and integration tests using Pytest (98%+ coverage) across multiple projects
- Introduced GitHub Actions for Continuous Integration, enabling automatic execution of tests for each pull request, reducing review time by 90%

PROJECT EXPERIENCE

Splitwiser – A Splitwise clone | Full Stack Development, Amazon Bedrock *June 2024 - December 2024*

- Created a Splitwise-inspired application featuring receipt scanning via camera or file upload, group and non-group expense tracking, and dynamic graphical and manual split functionalities
- Leveraged Amazon Bedrock API to convert receipt images into JSON data, to facilitate graphical expense visualization
- Utilized NextJS for UI and API development, Google Firebase for backend services, and Firestore for storing receipt data

Autokube | DevOps, CI/CD *August 2021 - March 2022*

- Led the development of the web application for automating Kubernetes cluster creation and modification
- Utilized MongoDB, Express.js, React, Node.js (MERN) stack for formulating APIs to initiate scripts, triggering a series of workflows for managing the backend Kubernetes clusters

EDUCATION

Binghamton University, State University of New York
Master of Science in Computer Science

December 2024

TECHNICAL SKILLS

Programming Languages and DB: C, Python, Java, JavaScript, TypeScript, MongoDB, PostgreSQL, MYSQL, Firebase
Frameworks and Software's: AWS, Amazon Bedrock, AWS Lambda, Dynamo DB, EC2, ExpressJS, ReactJS, NodeJS, Django, CI/CD, Pytest, Git, Linux, Kubernetes, TensorFlow, Keras, Pytorch, Pandas, Jupyter Notebook, Google Colab